Probability and Stochastic Processes

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Chapter 7 Stationary Processes

Many important processes have the property that their finite-dimensional distributions are invariant under time shifts (or space shifts). These processes are called *strict-sense stationary processes*. Another kind of stationary is *wide-sense stationary* while it has constant mean and its autocorrelation function is invariant under time shifts.

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Strict Stationary Processes

Definition

A ***strict-sense stationary process*** *(SSS) is a stochastic process* $\{X_t\}_{t\in T}$ *with the property that for any positive integer* n *and for any* $t_1, \cdots, t_n$ *and* $h \in T$*,*

$$(X_{t_1}, X_{t_2}, \cdots, X_{t_n}) \text{ and } (X_{t_1+h}, X_{t_2+h}, \cdots, X_{t_n+h})$$

have the same nth-order probability distribution function, i.e.,

$$F_X(x_1, \cdots, x_n; t_1, \cdots, t_n) = F_X(x_1, \cdots, x_n; t_1 + h, \cdots, t_n + h)$$

for all $x_1, \cdots, x_n \in \mathbb{R}$*.*

In general, the index set T can be $(-\infty, +\infty)$, $[0, +\infty)$, $\{0, \pm 1, \pm 2, \ldots\}$ or $\{0, 1, 2, \ldots\}$.
If T is discrete, then we say $\{X_t, t \in T\}$ is a stationary time sequence in the strict sense.

Strict Stationary Processes

Example

Suppose that $\{X_n\}_{n\geqslant 0}$ are IID random sequences, each of which has a uniform distribution in the interval $(0,1)$. Is $\{X_n\}_{n\geqslant 0}$ a stationary time sequence in the strict sense? Determine the value $E(X_n)$ and $E(X_nX_m), n, m = 0, 1, 2, \cdots$.

Solution. Suppose that the distribution function of X_n is $F(x)$. Then for all $k, h \in \mathbb{N}$ and $0 < n_1 < n_2 < \cdots < n_k$, the joint distribution functions of $(X_{n_1}, X_{n_2}, \ldots, X_{n_k})$ and $(X_{n_1+h}, X_{n_2+h}, \ldots, X_{n_k+h})$ are both

$$F(x_1, x_2, \ldots, x_k) = \prod_{i=1}^{k} F(x_k).$$

Strict Stationary Processes

It follows that $\{X_n\}_{n\geqslant 0}$ is a strict-sense stationary time sequence, and $E(X_n) = \frac{1}{2}$. If $m = n$, then

$$E(X_n X_m) = E(X_n^2) = \frac{1}{12} + \frac{1}{4} = \frac{1}{3}.$$

If $m \neq n$, then

$$E(X_n X_m) = E(X_n)E(X_m) = \frac{1}{4}.$$

Thus,

$$E(X_n X_m) = \begin{cases} \frac{1}{3} & \text{for } m = n, \\ \frac{1}{4} & \text{for } m \neq n. \end{cases}$$

$\square$

Strict Stationary Processes

Definition

A process $\{X_t\}_{t\in T}$ *is said to be of second order if*

$$E(X_t^2) < \infty \quad \text{for all } t \in T.$$

Theorem

Suppose that the process $\{X_t\}_{t\in T}$ *is strict-sense stationary and of second order. Then*
(i) *for all* $t \in T$, $E(X_t) =$ *constant;*
(ii) $E(X_s X_t)$ *depends only on the time difference* $t - s$ *for all* $t, s \in T$.

Strict Stationary Processes

Proof. (i) By the Cauchy-Schwarz inequality, we have

$$(E(X_t))^2 \leqslant E(X_t^2) < +\infty,$$

thus $E(X_t)$ exists.
If we let $h = -s$ in the definition of strict-sense stationary process, then we conclude that X_s and X_0 have the same distribution function. Hence, $E(X_s) = E(X_{s-s}) = E(X_0)$ is a constant.

Strict Stationary Processes

(ii) By the Cauchy-Schwarz inequality, we have

$$(E(X_s X_t))^2 \leqslant E(X_s^2)E(X_t^2) < +\infty.$$

Thus $E(X_s X_t)$ exists.
If we let $h = -s$, then (X_s, X_t) and (X_0, X_{t-s}) have the same distribution function. It follows that

$$E(X_s X_t) = E(X_0 X_{t-s}),$$

that is,

$$R_X(s, s+\tau) = E[X_0 X_\tau] = R_X(\tau),$$

where $\tau = t - s$. Therefore, $R_X(s,t)$ depends only on the time difference $t - s$. $\square$

Strict Stationary Processes

Furthermore,

$$C_X(t, t+\tau) = E[(X_t - \mu_X)(X_{t+\tau} - \mu_X)] = R_X(\tau) - \mu_X^2 = C_X(\tau)$$

depends only on τ and

$$\sigma_X^2 = C_X(0) = R_X(0) - \mu_X^2$$

is a constant.

Example

Suppose that Markov chain $\{X_n\}_{n \geqslant 0}$ with state space $S = \{0, 1, \cdots, N\}$ is ergodic and it has the unique invariant distribution π, started with X_0 distributed according to π. Is this Markov chain a strict-sense stationary process?

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Wide Stationary Processes

A weaker form of stationarity exists which does not directly constrain the nth-order probability density functions, but rather just the first- and second-order moments. This property, which is easier to check, is called wide-sense stationarity and will be quite useful.

Wide Stationary Processes

Definition

A process $\{X_t\}_{t\in T}$ *is said to be* ***wide-sense stationary*** *(WSS), if it is of second order and satisfies*
(i) *for all* $t \in T$, $E(X_t) = \mu_X$ =*constant;*
(ii) *for all* $t,\ t+\tau \in T$, $R_X(t, t+\tau) = R_X(\tau)$ *depends only on the time difference* τ.
In particular, a random sequence $\{X_n\}_{n\geqslant 0}$ *is said to be a stationary random sequence (or time sequence) in the wide sense if it is of second order and satisfies*
(i) *for all* $n \geqslant 0$, $E(X_n)$ =*constant,*
(ii) *for all* $m, n \geqslant 0, R_X(n, n+m) = R_X(m)$ *depends only on* m.

Wide Stationary Processes

Weak stationarity interests us more than strong stationarity for two reasons.

Firstly, the condition of strong stationarity is often too restrictive for certain applications;

secondly, many substantial and useful properties of stationary processes are derivable from weak stationarity alone.

Thus, the assertion that X is stationary should be interpreted to mean that X is weakly. Of course, there exist processes which are stationary but not strongly stationary and conversely processes without finite second moments may be strongly stationary but not weakly stationary.

Wide Stationary Processes

Example

Suppose that $\{X_n, n = 0, \pm 1, \pm 2, \dots\}$ is a pairwisely uncorrelated time sequence and $E(X_n) = 0$, $Var(X_n) = \sigma^2 > 0$. Then $\{X_n\}$ is wide-sense stationary.

Solution. Since $E(X_n) = 0$ and

$$E(X_m X_n) = \begin{cases} \sigma^2 & \text{for } m = n, \\ 0 & \text{for } m \neq n, \end{cases}$$

$\{X_n\}$ is a wide-sense stationary time sequence. $\square$

Wide Stationary Processes

Recall a sinusoid with random phase $X_t = a\sin(\omega t + \Theta)$, the mean and the autocorrelation function of X_t are $\mu_X(t) = 0$ and $R_X(t, t+\tau) = \frac{a^2}{2}\cos\tau$ respectively. Thus X_t is wide-sense stationary. In fact we have a general results as follows.

Example

Suppose that S_t is a deterministic function with period T and the random variable Θ has a uniform distribution in the interval $(0, T)$. Let $X_t = S_{t+\Theta}$. Discuss the stationarity of X_t.

Wide Stationary Processes

Example

Let X_t be strict-sense stationary with finite second moment, and let $Y_t = a + bt + X_t$, $W_t = t[X_t - E(X_t)]$ and

$$Z_t = \begin{cases} \alpha + X_t & \text{if } t \leqslant t_0, \\ \beta + X_t & \text{if } t > t_0. \end{cases}$$

Are Y_t, W_t and Z_t stationary?

Wide Stationary Processes

Example

(**Random Telegraph Signal**) We are given a stochastic process X_t taking only the values I and $-I$ with equal probability

$$P(X_t = I) = P(X_t = -I) = \frac{1}{2}.$$

Let $N(t, t+l)$ be the times of the sign changing in the interval $(t, t+l)$. Suppose that $N(t, t+l)$ has a Poisson distribution with mean λl, that is, if we set

$$A_k := \{N(t, t+l) = k\},$$

then

$$P(A_k) = \frac{(\lambda l)^k}{k!} e^{-\lambda l}, \ k = 0, 1, \cdots,$$

where $\lambda > 0$ denotes the average times of the sign changing in the unit time. Discuss the stationarity of X_t.

Wide Stationary Processes

An important class of wide-sense stationary processes is white noise, which have zero mean and are serially uncorrelated.

Definition

(**White noise processes**) *A wide-sense stationary process W_t is white noise if*

$$E(W_t) = 0 \quad \text{and} \quad Cov(W_t, W_{t-j}) = 0 \text{ for all } j \neq 0.$$

An independent and identically distributed sequence with mean 0 and finite variance is called an independent white noise process.

Wide Stationary Processes

Example

Let X be a random variable uniformly distributed in the interval $(0, 2\pi)$. Define $W_t = cos(tX)(t = 1, 2, \cdots)$. It can be shown that $E(W_t) = 0, Var(W_t) = 1/2$ and $Cov(W_t, W_s) = 0$ for $t \neq s$. So W_t is white noise. However, the values are clearly dependent, so the sequence is not independent white noise. Furthermore, the process is not strictly stationary. (Suppose that $tX = 2\pi$ vs. $tX = \pi/4$. The joint distribution of two observations will depend on their relative placement in the sequence.)

Wide Stationary Processes

Let X_t be a WSS process and relabel $R_X(t, t+\tau)$ as $R_X(\tau)$. We can see $R_X(\tau)$ has the following properties.

Property

(i) *$R_X(\tau)$ is real and even, i.e., $R_X(-\tau) = R_X(\tau)$ for all τ.*
(ii) $|R_X(\tau)| \leqslant R_X(0) = E(X_t^2)$.

Property

If X_t is a periodic process, then its autocorrelation function is periodic with period T. That is, $R_X(\tau + T) = R_X(\tau)$.

Wide Stationary Processes

Property

For all $n \in \mathbb{N}$, *for all* $t_1 < t_2 < \cdots < t_n$, *and for all constants* $a_1, a_2, \ldots, a_n \in \mathbb{R}$, *we have*

$$\sum_{j=1}^{n} \sum_{k=1}^{n} R_X(t_j - t_k) a_j a_k \geqslant 0.$$

Remark

If $R_X(\tau)$ *drops quickly with* τ, *then samples become uncorrelated quickly as we increase* τ. *Conversely, if* $R_X(\tau)$ *drops slowly with* τ, *then samples are highly correlated. So* $R_X(\tau)$ *is a measure of the rate of change of* X_t *with time* t, *i.e., "the frequency response of* X_t*".*

Wide Stationary Processes

Theorem

If X_t is a WSS process, then it is mean square continuous if and only if its autocorrelation function $R_X(\tau)$ is continuous for $\tau = 0$.

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Joint Stationary Processes

Definition

Suppose that $\{X_t\}$ *and* $\{Y_t\}$ *are two stationary processes. If for all* $t, t+\tau \in T$,

$$R_{XY}(t, t+\tau) = E(X_t Y_{t+\tau}) = R_{XY}(\tau),$$

then we say that $\{X_t\}$ *and* $\{Y_t\}$ *are jointly stationary processes and* $R_{XY}(\tau)$ *is called their cross-correlation function.*

Joint Stationary Processes

Example

Assume U and V are independent normally distributed random variables having zero mean and unit variance. Let T be the circumference of the unit circle, represented by $T = [0, 2\pi]$. For $t \in T$, define the bivariate process (Y_t, Z_t), where

$$Y_t = U \sin t + V \cos t,$$
$$Z_t = -U \cos t + V \sin t.$$

Prove that (Y_t, Z_t) is a joint stationary process.

Joint Stationary Processes

The cross-correlation function $R_{XY}(\tau)$ has the following properties.

Property

(i) $R_{XY}(0) = R_{YX}(0)$;
(ii) $R_{XY}(\tau) = R_{YX}(-\tau)$;
(iii) $|R_{XY}(\tau)|^2 \leqslant R_X(0)R_Y(0)$;
(iv) $|R_{XY}(\tau)| \leqslant \frac{1}{2}[R_X(0) + R_Y(0)]$.

Joint Stationary Processes

Definition

A complex valued process $Z_t = X_t + iY_t$ *is said to be a wide-sense stationary process if it satisfies*
(i) $\mu_Z(t) = \mu_X + i\mu_Y = constant$,
(ii) $R_Z(t, t+\tau) = R_Z(\tau)$ *depends only on* τ.

Remark

(i) *If* $\{X_t\}$ *and* $\{Y_t\}$ *are two real-valued, jointly stationary processes, then* $Z_t = X_t + iY_t$ *is a complex valued, stationary process.*
(ii) $R_Z(\tau) = \overline{R_Z(-\tau)}$.

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Ergodicity of Stationary Process

If the random process is such that the behavior of almost every particular realization over time is representative of the behavior down the ensemble, then the process is called ergodic. It is often said that ensemble averaging and time averaging are identical for a statistically-stationary system, but are different for a statistically-nonstationary system.

Ergodicity of Stationary Process

An ensemble average is a convenient theoretical concept since it is directly related to the probability density functions, which can be generally obtained by the theoretical analysis of a given physical system. For example, X_t is a WSS process,

$$\mu_X = \mu_X(t) = \int_{-\infty}^{\infty} x f(x,t) dx = \int_{-\infty}^{\infty} x f(x) dx,$$

$$R_X(\tau) = R_X(t, t+\tau) = \iint_{\mathbb{R}^2} xy f(x, y; t, t+\tau) dx dy$$

$$= \iint_{\mathbb{R}^2} xy f(x, y; \tau) dx dy.$$

Ergodicity of Stationary Process

In many cases, the information of probability density functions is unknown. On the other hand, we can use the law of large numbers to obtain ensemble average. That means we need prepare an enough number of identical systems and estimate the real values by

$$\mu_X \approx \frac{1}{N}\sum_{k=1}^{n} x_t^{(k)} \quad \text{and} \quad R_X(\tau) \approx \frac{1}{N}\sum_{k=1}^{n} x_t^{(k)} x_{t+\tau}^{(k)},$$

for a given t, where $x_t^{(k)}$ is a sample path of the process X_t, which usually cannot be achieved in a real situation. For instance, $x(t)$ might be a record of tidal heights as a function of time over particular 20 years, or perhaps of atmosphere turbulence over a certain week at a given location. Of course, there is no possibility of repeating these experiments under the same conditions.

Ergodicity of Stationary Process

What we usually use is the time average

$$\frac{1}{2T}\int_{-T}^{T} x(t)dt \quad \text{and} \quad \frac{1}{2T}\int_{-T}^{T} x(t)x(t+\tau)dt.$$

The concept of ergodicity is sophisticated and subtle, but the essential idea is described here. In this section, we will present two ergodic theorems for stationary processes. One is in mean, and another is in correlation. Just as there are strong and weak laws of large numbers, there are a variety of ergodic theorems, differing in their assumptions and in the modes of convergence. Here the natural mode of convergence is in mean square.

Ergodicity of Stationary Process

Definition

The time average of a WSS stochastic process $\{X_t\}_{t\in\mathbb{R}}$ *is defined by*

$$\langle X_t \rangle = \text{l.i.m.}_{T\to+\infty} \frac{1}{2T} \int_{-T}^{T} X_t dt.$$

Definition

The time correlation function of a WSS stochastic process $\{X_t\}_{t\in\mathbb{R}}$ *is defined by*

$$\langle X_t X_{t+\tau} \rangle = \text{l.i.m.}_{T\to+\infty} \frac{1}{2T} \int_{-T}^{T} X_t X_{t+\tau} dt.$$

Ergodicity of Stationary Process

Definition

A WSS stochastic process $\{X_t\}_{t\in\mathbb{R}}$ is ergodic in the mean if the time average converges to the ensemble average μ_X, that is,

$$\lim_{T\to\infty} E\left[\left(\frac{1}{2T}\int_{-T}^{T} X_t \mathsf{d}t - \mu_X\right)^2\right] = 0. \tag{1}$$

Definition

A WSS stochastic process $\{X_t\}_{t\in\mathbb{R}}$ is ergodic in correlation for shift (lag) τ if

$$\lim_{T\to\infty} E\left[\left(\frac{1}{2T}\int_{-T}^{T} X_t X_{t+\tau} dt - R_X(\tau)\right)^2\right] = 0. \tag{2}$$

If this condition is true for all τ, we say X_t is ergodic in correlation.

Ergodicity of Stationary Process

Definition

Suppose that $\{X_t\}_{t\in\mathbb{R}}$ is a WSS process. If it is ergodic in the mean and the correlation, then we say it is an ergodic process.

Example

Consider the WSS process

$$X_t = \sin(\omega_0 t + \Theta),\ t \in (-\infty, +\infty),$$

where ω_0 is a constant and $\Theta \sim U(0, 2\pi)$. Discuss the ergodicity of this process.

Ergodicity of Stationary Process

Example

Suppose that the stationary process $X_t = Y$, where the random variable

$$Y = \begin{cases} 1 & \text{with probability } \frac{1}{2}, \\ -1 & \text{with probability } \frac{1}{2}. \end{cases}$$

Discuss the ergodicity of $\{X_t\}$.

Ergodicity of Stationary Process

Example

Consider a basket full of batteries. There are some flashlight batteries, some car batteries, and several other kinds of batteries. Suppose that a battery is selected at random and its voltage is measured. This battery voltage U_t is a member function selected from a certain sub-group of constant battery voltages. This process is stationary but not ergodic in the mean or correlation. The time average is equal to the particular battery voltage selected. The statistical average is some other number, depending on what is in the basket. Thus equation (1) does not hold.

Ergodicity of Stationary Process

Theorem

A mean-square continuous and WSS stochastic process X_t is ergodic in the mean iff its covariance function, that is, $R_X(\tau) - \mu_X^2$ satisfies

$$\lim_{T \to +\infty} \frac{1}{T} \int_0^{2T} \left(1 - \frac{\tau}{2T}\right) \left[R_X(\tau) - \mu_X^2\right] d\tau = 0. \tag{3}$$

Corralary

A mean-square continuous and WSS stochastic process $\{X_t\}$ is ergodic in the mean iff

$$\lim_{T \to +\infty} \frac{1}{T} \int_0^{2T} (1 - \frac{\tau}{2T}) R_X(\tau) d\tau = \mu_X^2.$$

Ergodicity of Stationary Process

Corralary

Suppose that $\{X_t\}$ *is a mean-square continuous and WSS stochastic process. If*

$$\int_{-\infty}^{+\infty} |R_X(\tau)| d\tau < \infty,$$

then it is ergodic in the mean iff $\mu_X = 0$.

Example

Consider the random telegraph signal $\{X_t\}$ with the mean function $\mu_X = 0$ and the autocorrelation function $R_X(\tau) = I^2 e^{-2\lambda|\tau|}$. Is $\{X_t\}$ ergodic in the mean?

Ergodicity of Stationary Process

Theorem

Suppose that $\{X_t\}$ *is a mean-square continuous and WSS stochastic process, and for each* τ, $\{X_t X_{t+\tau}\}$ *is also a mean-square continuous and WSS stochastic process. Then* $\{X_t\}$ *is ergodic in correlation iff*

$$\lim_{T\to+\infty} \frac{1}{T} \int_0^{2T} \left(1 - \frac{\tau_1}{2T}\right) \left[B(\tau_1) - R_X^2(\tau)\right] d\tau_1 = 0,$$

where $B(\tau_1) = E(X_t X_{t+\tau} X_{t+\tau_1} X_{t+\tau+\tau_1})$.

Ergodicity of Stationary Process

If the index set T of the stationary process we considered is $[0, \infty)$, then we have

Theorem

A mean-square continuous and WSS stochastic process X_t is ergodic in the mean iff its covariance function, that is, $R_X(\tau) - \mu_X^2$ satisfies

$$\lim_{T \to +\infty} \frac{1}{T} \int_0^T \left(1 - \frac{\tau}{T}\right) \left[R_X(\tau) - \mu_X^2\right] d\tau = 0.$$

$\{X_t\}$ is ergodic in correlation iff

$$\lim_{T \to +\infty} \frac{1}{T} \int_0^T \left(1 - \frac{\tau_1}{T}\right) \left[B(\tau_1) - R_X^2(\tau)\right] d\tau_1 = 0,$$

where $B(\tau_1) = E(X_t X_{t+\tau} X_{t+\tau_1} X_{t+\tau+\tau_1})$.

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Power Spectral Density of Stationary Processes

If x_t is the voltage across a $R = 1W$ resistor, then the total energy in x_t is

$$E = \int_{-\infty}^{+\infty} x_t^2 dt.$$

But for communications signals, the energy is effectively infinite (the signals are of unlimited duration), so we usually work with Power quantities. When the voltage is random, how about its power? In this section, we will introduce the definitions of the average power and power spectral density of a stationary process. The properties of power spectral density are also reached.

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Average Power and Power Spectral Density

Theorem

Suppose that x_t $(t \in \mathbb{R})$ is a deterministic real-valued function satisfying the Dirichlet condition. Then
(i) *the Fourier transform of x_t exists and we can compute it by the integral*

$$F(\omega) = \int_{-\infty}^{+\infty} x_t e^{-i\omega t} dt.$$

The Fourier transform $F(\omega)$ of x_t is often called frequency spectrum.
(ii) *The inverse Fourier transform exists, which equals x_t at all points of continuity,*

$$x_t = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{i\omega t} d\omega.$$

Average Power and Power Spectral Density

Theorem

(i) *$F(\omega)$ is a complex-valued function and*

$$F(-\omega) = \int_{-\infty}^{+\infty} x_t e^{i\omega t} dt = \overline{F(\omega)}.$$

(ii) *The Parseval equality holds, that is,*

$$\int_{-\infty}^{+\infty} x^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |F(\omega)|^2 d\omega.$$

$|F(\omega)|^2$ denotes the distribution of the total energy of x_t and it is often called the **energy spectral density**. Thus the Parseval equality can be seen as the spectral representation of the total energy.

Average Power and Power Spectral Density

In many cases, the total energy does not exists. Then we always need compute the average power of x_t. The standard steps to find the average power are as follows.

S1. Given a function x_t ($t \in \mathbb{R}$). Consider the finite support segment,

$$x_T(t) = \begin{cases} x_t & \text{if } |t| \leqslant T, \\ 0 & \text{if } |t| > T. \end{cases}$$

S2. Compute the Fourier transform of $x_T(t)$

$$F_T(\omega) = \int_{-\infty}^{+\infty} x_T(t) e^{-i\omega t} dt = \int_{-T}^{T} x_T(t) e^{-i\omega t} dt.$$

S3. The Parseval equality holds, that is,

$$\int_{-\infty}^{+\infty} x_T^2(t) \mathsf{d}t = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |F_T(\omega)|^2 d\omega.$$

Average Power and Power Spectral Density

S4. Dividing by $2T$, we get

$$\frac{1}{2T}\int_{-T}^{T} x_t^2 dt = \frac{1}{2T}\int_{-\infty}^{+\infty} x_T^2(t)dt = \frac{1}{4\pi T}\int_{-\infty}^{+\infty} |F_T(\omega)|^2 d\omega.$$

S5. Let $T \to +\infty$. We get the average power

$$Q = \lim_{T\to+\infty}\frac{1}{2T}\int_{-T}^{T} x^2(t)dt = \frac{1}{2\pi}\int_{-\infty}^{+\infty}\lim_{T\to+\infty}\frac{1}{2T}|F_T(\omega)|^2 d\omega. \tag{4}$$

Denote the integrand in the right hand of equation (4) by $S_X(\omega)$, i.e.,

$$S_X(\omega) = \lim_{T\to+\infty}\frac{1}{2T}|F_T(\omega)|^2,$$

which is called the **power spectral density**(PSD) of x_t.

Average Power and Power Spectral Density

When the process X_t we meet is random, how about the average power and the corresponding power spectral density?

In fact, for deterministic process, we only consider the time average in equation (4). Fortunately, for random process X_t, our earlier results can be extended to cover X_t by the inclusion of an extra averaging or expectation step over all possible values of X_t since the integral in equation (4) will be a mean-square integral.

Average Power and Power Spectral Density

Assume X_t is a mean-square continuous stationary process. Similarly to above steps, we construct a truncated function $X_T(t)$, compute the Fourier transform of $X_T(t)$, find the total energy for $X_T(t)$, dividing by $2T$, find the expectations of both sides and let $T \to +\infty$, then we get the average power

$$\lim_{T\to+\infty} E\left(\frac{1}{2T}\int_{-T}^{T} X_t^2 dt\right) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \lim_{T\to+\infty} \frac{1}{2T} E\big(|F_T(\omega)|^2\big) d\omega. \quad (5)$$

Average Power and Power Spectral Density

Definition

Suppose that X_t is a mean-square continuous stationary process.

$$Q := \lim_{T \to +\infty} E\left(\frac{1}{2T}\int_{-T}^{T} X_t^2 dt\right)$$

is defined as the average power of the process X_t. The power spectral density of X_t is defined to be

$$S_X(\omega) = \lim_{T \to +\infty} \frac{1}{2T} E\big(|F_T(\omega)|^2\big).$$

Average Power and Power Spectral Density

The power spectral density for a signal is a measure of its power distribution as a function of frequency. It is a useful concept which allows us to determine the bandwidth required of a transmission system.
Since X_t is stationary, we have

$$Q = \lim_{T \to +\infty} E\left(\frac{1}{2T}\int_{-T}^{T} X_t^2 dt\right) = \lim_{T \to +\infty} \frac{1}{2T}\int_{-T}^{T} E(X_t^2) dt = R_X(0). \tag{6}$$

Average Power and Power Spectral Density

Example

Find the average power for the sine wave with random phase

$$X_t = a\sin(\omega_0 t + \Theta),$$

where a and ω_0 are constants and Θ has a uniform distribution in the interval $(0, 2\pi)$.

Solution. Since $R_X(\tau) = \frac{a^2}{2}\cos\tau$, we have the average power

$$Q = R_X(0) = \frac{a^2}{2}.$$

$\square$

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Power Spectral Density and Autocorrelation Function

In this section, we introduce some properties of power spectral density and autocorrelation function of stationary process.
From the definition of power spectral density, we know

Property

(i) $S_X(\omega)$ *is nonnegative for all* ω.
(ii) $S_X(\omega)$ *is a real and even function, that is* $\overline{S}_X(\omega) = S_X(\omega)$ *and* $S_X(\omega) = S_X(-\omega)$.
(iii) *If* $S_X(\omega)$ *is a rational function, then it must have the following form:*

$$S_X(\omega) = S_0 \frac{\omega^{2n} + a_{2n-2}\omega^{2n-2} + \cdots + a_0}{\omega^{2m} + b_{2m-2}\omega^{2m-2} + \cdots + b_0},$$

where $a_{2n-i}, b_{2n-j} \quad (i = 2, \cdots, 2n,\ j = 2, 4, \cdots, 2m)$ *are constants and* $S_0 > 0$, $m > n$, *and the denominator has no real root.*

Power Spectral Density and Autocorrelation Function

Theorem

(**Wiener-Khintchine theorem**) *Suppose that $R_X(\tau)$ is the autocorrelation function of the wide-sense stationary process $\{X_t\}$ and $S_X(\omega)$ is the power spectral density. If*

$$\int_{-\infty}^{+\infty} |R_X(\tau)| d\tau < +\infty, \tag{7}$$

then

(i)

$$S_X(\omega) = \int_{-\infty}^{+\infty} R_X(\tau) e^{-i\omega\tau} d\tau. \tag{8}$$

(ii) *Under condition (7), the inverse Fourier transform is given by*

$$R_X(\tau) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_X(\omega) e^{i\omega\tau} d\omega. \tag{9}$$

Power Spectral Density and Autocorrelation Function

Remark

$R_X(\tau)$ is the frequency response of X_t. Its Fourier Transformation is just the average power density over frequency.
Equations (8) and (9) illustrate the relationship between the property in the time domain and the property in the frequency domain of the stationary process. Using equation (9) we find

$$R_X(0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_X(\omega) d\omega. \tag{10}$$

Power Spectral Density and Autocorrelation Function

Corralary

If $R_X(\tau)$ and $S_X(\omega)$ are both even, then we have the following equalities

$$S_X(\omega) = 2\int_0^{+\infty} R_X(\tau)\cos\omega\tau d\tau, \tag{11}$$

$$R_X(\tau) = \frac{1}{\pi}\int_0^{+\infty} S_X(\omega)\cos\omega\tau d\omega. \tag{12}$$

Example

Find the power spectral density for the following exponential autocorrelation function $R_X(\tau) = Ae^{-\beta|\tau|}$ with parameters $A > 0$ and $\beta > 0$.

Power Spectral Density and Autocorrelation Function

Example

Assume that the power spectral density of stochastic process X_t is

$$S_X(\omega) = \frac{\omega^2 + 4}{\omega^4 + 10\omega^2 + 9}.$$

Find the corresponding autocorrelation function $R_X(\tau)$.

Example

Let $\{X_t\}$ be a stationary process with autocorrelation function

$$R_X(\tau) = e^{-a|\tau|} \cos \omega_0 \tau,$$

where $a > 0$ and ω_0 are constants. Find the power spectral density $S_X(\omega)$.

Power Spectral Density and Autocorrelation Function

Example

Find the power spectral density for the following stochastic process $X_t = a\cos(\omega_0 t + \Theta)$, where a and ω_0 are constants and Θ has a uniform distribution in the interval $(0, 2\pi)$.

Example

(**Power spectral density of white noise**) The autocorrelation function of a white noise process $\{W_t\}_{t\in\mathbb{R}}$ with parameter σ^2 is given by $R_W(\tau) = \sigma^2\delta(\tau)$. Hence the power spectral density, its Fourier transform, is just

$$S_W(\omega) = \sigma^2, \qquad -\infty < \omega < +\infty.$$

Power Spectral Density and Autocorrelation Function

Theorem

Suppose that $\{X_n\}$ *is a stationary random sequence with the mean function* $E(X_n) = 0$ *and the autocorrelation function* $R_X(m) = E(X_n X_{n+m})$. *If*

$$\sum_{m=-\infty}^{+\infty} |R_X(m)| < +\infty,$$

then the power spectral density of $\{X_n\}$ *is the Fourier transform of* $R_X(m)$, *that is,*

$$S_X(\omega) = \sum_{m=-\infty}^{+\infty} R_X(m) e^{-im\omega}, \quad for - \pi \leqslant \omega \leqslant \pi.$$

The inverse Fourier transform is given as

$$R_X(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_X(\omega) e^{im\omega} d\omega.$$

Power Spectral Density and Autocorrelation Function

Example

Suppose that $\{X_n, n = 0, \pm 1, \pm 2, \ldots,\}$ is a stationary random sequence with the mean function 0 and the autocorrelation function

$$R_X(n) = \begin{cases} \sigma^2 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0. \end{cases}$$

Find the power spectral density $S_X(\omega)$.

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Cross-Power Spectral Density

Definition

Let $\{X_t\}$ and $\{Y_t\}$ be two real-valued, jointly stationary processes. Then we can define the cross-power spectral density,

$$S_{XY}(\omega) = \lim_{T\to+\infty} \frac{1}{2T} E\left[F_T(-\omega)G_T(\omega)\right],$$

$$S_{YX}(\omega) = \lim_{T\to+\infty} \frac{1}{2T} E\left[G_T(-\omega)F_T(\omega)\right],$$

where $F_T(\omega)$ is the Fourier transform of the finite support segment $X_T(t)$ of $\{X_t\}$ and $G_T(\omega)$ is the Fourier transform of the finite support segment $Y_T(t)$ of $\{Y_t\}$.

Cross-Power Spectral Density

Theorem

If the cross-correlation function $R_{XY}(\tau)$ satisfies

$$\int_{-\infty}^{+\infty} |R_{XY}(\tau)| d\tau < +\infty,$$

then the cross-power spectral density $S_{XY}(\omega)$ is the Fourier transform of $R_{XY}(\tau)$, that is,

$$S_{XY}(\omega) = \int_{-\infty}^{+\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau. \tag{13}$$

The inverse Fourier transform is

$$R_{XY}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_{XY}(\omega) e^{i\omega\tau} d\omega. \tag{14}$$

Cross-Power Spectral Density

Property

(i) $S_{XY}(\omega)$ *is* <u>*not a real*</u>*-valued, nonnegative and even function.*
(ii) $S_{XY}(\omega) = \overline{S_{YX}(\omega)} = S_{YX}(-\omega)$.
(iii) $Re[S_{XY}(\omega)]$ *and* $Re[S_{YX}(\omega)]$ *are even functions of* ω,
(iv) $Im[S_{XY}(\omega)]$ *and* $Im[S_{YX}(\omega)]$ *are odd functions of* ω.
(v) *If the processes* $\{X_t\}$ *and* $\{Y_t\}$ *are orthogonal, then* $R_{XY}(s,t) = 0$, $S_{XY}(\omega) = 0$ *and* $S_{YX}(\omega) = 0$.

Cross-Power Spectral Density

(vi) Let $\{X_t\}$ and $\{Y_t\}$ be two jointly stationary processes with the constant mean functions $\mu_X \neq 0$ and $\mu_Y \neq 0$ respectively. If they are uncorrelated, then

$$S_{XY}(\omega) = S_{YX}(\omega) = 2\pi\mu_X\mu_Y\delta(\omega).$$

(vii) Suppose that $\{X_t\}$ and $\{Y_t\}$ are two jointly stationary processes. Then

$$|S_{XY}(\omega)|^2 \leqslant S_X(\omega)S_Y(\omega).$$

Cross-Power Spectral Density

Example

Suppose that

$$X_t = A\sin(\omega_0 t + \Theta) \quad \text{and} \quad Y_t = B\sin(\omega_0 t - \varphi + \Theta),$$

where A, B, ω, φ are constants, and Θ is uniform in the interval $(0, 2\pi)$.
(a) Are X_t and Y_t joint stationary?
(b) If X_t and Y_t are joint stationary, find their cross-power spectral density $S_{XY}(\omega)$.

Cross-Power Spectral Density

Example

Let $\{X_t\}$ and $\{Y_t\}$ be two jointly stationary processes. The cross-power spectral density is given by

$$S_{XY}(\omega) = \begin{cases} a + i\frac{b\omega}{\Omega} & \text{for} - \Omega < \omega < \Omega, \\ 0 & \text{otherwise,} \end{cases}$$

where $\Omega > 0$, a and b are constants. Calculate the cross-correlation function $R_{XY}(\tau)$.

Cross-Power Spectral Density

The cross-power spectral density is usually used to study the sum of two joint stationary processes. Sometimes it can be used to describe the relationship between the input process and output process.

Chapter 7 Stationary Processes

- Section 7.1 Stationary Processes
- Section 7.2 Ergodicity of Stationary Process
- Section 7.3 Power Spectral Density of Stationary Processes
- Section 7.4 Stationary Processes and Linear Systems

Stationary Processes and Linear Systems

In this section, we discuss the analysis of linear systems with random processes as inputs. To begin with, we review some concepts from linear system theory for deterministic inputs. Then, we extend these concepts to systems with stochastic processes as inputs.

Definition

Assume $x_1(t)$ and $x_2(t)$ are two deterministic time functions, α and β are two scalar constants. Let the linear system be described by the operator equation $y = L[x]$. The system is said to be linear if

$$L[\alpha x_1(t) + \beta x_2(t)] = \alpha L[x_1(t)] + \beta L[x_2(t)] = \alpha y_1(t) + \beta y_2(t)$$

for all admissible functions $x_1(t)$ and $x_2(t)$ and all scalars α and β.

Stationary Processes and Linear Systems

A general linear system with input $x(t)$ and output $y(t)$ has the form

$$y(t) = \int_{-\infty}^{+\infty} h(t,\tau)x(\tau)d\tau,$$

where $h(t,\tau)$ is referred to as the impulse response or weighting function of the system.
The system is said to be time-invariant if
$h(t,\tau) = h(t-\tau,0) \equiv h(t-\tau)$. If the system is time-invariant, then $y(t)$ is the convolution of $h(t)$ and $x(t)$. That is,

$$y(t) = \int_{-\infty}^{+\infty} h(t-\tau)x(\tau)d\tau = \int_{-\infty}^{+\infty} h(\tau)x(t-\tau)d\tau.$$

Stationary Processes and Linear Systems

Definition

We say that a linear system L is stationary if the impulse response $h(t)$ satisfies:

$$\int_{-\infty}^{+\infty} |h(t)| dt < +\infty.$$

If we input the stationary stochastis process X_t to a linear stationary system with impulse response $h(t)$, then the output stochastic process Y_t can be expressed as the convolution integral,

$$Y_t = \int_{-\infty}^{+\infty} h(\tau) X_{t-\tau} d\tau = \int_{-\infty}^{+\infty} X_\tau h(t-\tau) d\tau, \qquad (15)$$

when this integral exists.

Stationary Processes and Linear Systems

Theorem

If we input a stationary stochastic process $\{X_t\}$ with mean function μ_X and autocorrelation function $R_X(\tau)$ to a linear stationary system with impulse response $h(t)$, then the output stochastic process $\{Y_t\}$ is also stationary. Furthermore, $\{X_t\}$ and $\{Y_t\}$ are jointly stationary processes, and

$$\mu_Y = \mu_X \int_{-\infty}^{+\infty} h(t)dt, \tag{16}$$

$$R_Y(\tau) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(\tau_1)h(\tau_2)R_X(\tau + \tau_1 - \tau_2)d\tau_1 \mathsf{d}\tau_2, \tag{17}$$

$$R_{XY}(\tau) = \int_{-\infty}^{+\infty} R_X(\lambda)h(\tau - \lambda)d\lambda. \tag{18}$$

Stationary Processes and Linear Systems

In fact, $\int_{-\infty}^{+\infty} h(\tau)d\tau$ is the system's frequency response $H(\omega)$, i.e.,

$$H(\omega) = \int_{-\infty}^{+\infty} h(\tau)d\tau.$$

Then by using the equation (**??**), we see $\mu_Y = \mu_X H(\omega)$. From equations (**??**) and (**??**), we have

$$\begin{aligned}
R_{XY}(\tau) &= R_X(\tau) * h(\tau). \\
R_Y(\tau) &= R_X(\tau) * h(-\tau) * h(\tau) \\
&= R_X(\tau) * h(\tau) * h(-\tau) = R_{XY}(\tau) * h(-\tau).
\end{aligned}$$

Stationary Processes and Linear Systems

Example

The autocorrelation function of a white noise process X_t is given by $R_X(\tau) = \frac{N_0}{2}\delta(\tau)$. The process X_t is now input to the linear system with impulse response $h(t)$ that is given by

$$\alpha = \frac{1}{RC}, \quad h(t) = \begin{cases} \alpha e^{-\alpha t} & \text{for } t \geqslant 0, \\ 0 & \text{for } t < 0. \end{cases}$$

(a) Find the autocorrelation function of the output.
(b) Find the average power of the output.
(c) Find the cross-correlation function between the input process and the output process.

Stationary Processes and Linear Systems

Example

Consider the linear time-invariant system L whose input is the stochastic process $X_t = e^{i\omega_0 t}$. The frequency response of the system is $H(\omega)$. Find the output process Y_t.

Theorem

Let the stationary process $\{X_t\}$ with mean μ_X and autocorrelation $R_X(\tau)$ be input to the filter with output $\{Y_t\}$. The power spectral density of $\{X_t\}$ is $S_X(\omega)$ and the system's frequency response is $H(\omega)$. Then the power spectral density of $\{Y_t\}$ is

$$S_Y(\omega) = |H(\omega)|^2 S_X(\omega). \tag{19}$$

The cross-power spectral density of $\{X_t\}$ and $\{Y_t\}$ is

$$S_{XY}(\omega) = H(\omega) S_X(\omega).$$

Stationary Processes and Linear Systems

Example

Suppose that the input process $\{X_t\}$ is a white noise process with the power spectral density $S_X(\omega) = \frac{N_0}{2}$. The frequency response and the impulse response are given by

$$H(\omega) = \frac{i\omega RC}{1 + i\omega RC}, \quad h(t) = \delta(t) - \frac{1}{RC} e^{-\frac{t}{RC}}.$$

Find the autocorrelation function of the output process $\{Y_t\}$.

Summary

- A second order process $\{X_t\}_{t\in T}$ is wide-sense stationary (WSS), if it satisfies
  (i) for all $t \in T$, $E(X_t) = \mu_X$=constant;
  (ii) for all $t,\ t+\tau \in T$, $E(X_t X_{t+\tau}) = R_X(\tau)$ depends only on the time difference τ.
- The properties of the autocorrelation function $R_X(\tau)$ of a WSS process X_t.
  (i) $R_X(\tau)$ is real and even, i.e. $R_X(-\tau) = R_X(\tau)$ for all τ.
  (ii) $|R_X(\tau)| \leqslant R_X(0) = E[X^2(t)]$.
- Suppose that $\{X_t\}$ and $\{Y_t\}$ are two stationary processes. If for all $t, t+\tau \in T$,

$$R_{XY}(t, t+\tau) = E(X_t Y_{t+\tau}) = R_{XY}(\tau),$$

  then we say that $\{X_t\}$ and $\{Y_t\}$ are jointly stationary processes and $R_{XY}(\tau)$ is called their cross-correlation function.

Summary

- Assume X_t is stationary. Then

$$Q = \lim_{T\to+\infty} E\left(\frac{1}{2T}\int_{-T}^{T} X_t^2 dt\right) = \lim_{T\to+\infty} \frac{1}{2T}\int_{-T}^{T} E(X_t^2)dt = R_X(0)$$

- (**Wiener-Khintchine theorem**) If $\int_{-\infty}^{+\infty} |R_X(\tau)|d\tau < +\infty$, then

$$S_X(\omega) = \int_{-\infty}^{+\infty} R_X(\tau)e^{-i\omega\tau}d\tau.$$

$$R_X(\tau) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} S_X(\omega)e^{i\omega\tau}d\omega.$$

Stationary Processes and Linear Systems

- If we input a stationary stochastic process $\{X(t)\}$ with mean function μ_X and autocorrelation function $R_X(\tau)$ to a linear stationary system, then the output stochastic process $\{Y(t)\}$ is also stationary. Furthermore, $\{X(t)\}$ and $\{Y(t)\}$ are jointly stationary processes. We have

$$\mu_Y(t) = \mu_X \int_{-\infty}^{+\infty} h(t)dt,$$

$$R_Y(\tau) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} h(\tau_1)h(\tau_2)R_X(\tau+\tau_1-\tau_2)d\tau_1 \mathsf{d}\tau_2,$$

$$R_{XY}(\tau) = \int_{-\infty}^{+\infty} R_X(\lambda)h(\tau-\lambda)d\lambda,$$

$$S_Y(\omega) = |H(\omega)|^2 S_X(\omega),$$

$$S_{XY}(\omega) = H(\omega)S_X(\omega).$$

The end

Thank you !